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Treatment guideline for adult patients with bloodstream infections

Purpose:
This guideline is intended to help guide antimicrobial therapy for patients admitted to adult service lines following the results of Gram stain, organism identification (with or without Biofire™ BCID2 molecular resistance results), and antimicrobial susceptibilities. Deviation from the recommendations in this guideline may be required for patients with concomitant infections, history of resistant pathogens, or with antimicrobial allergies or intolerance. The recommendations in this guideline reflect susceptibility patterns found at Michigan Medicine.

BCID2 is able to provide identification of the following organisms and is available within ~2 hours from the time of Gram stain on a positive blood culture bottle:

Gram positive[a]	Gram negative[d]	Fungus
Enterococcus faecalis	*Acinetobacter calcoaceticus-baumannii* complex	*Candida albicans*
Enterococcus faecium	*Bacteroides fragilis*	*Candida auris*
Listeria monocytogenes	*Psudomonas aeruginosa*	*Candida glabrata (Nakaseomyces glabrata)*
Staphylococcus aureus	*Stenotrophomonas maltophilia*	*Candida krusei (Pichia kudriavzevii)*
Staphylococcus epidermidids	*Haemophilus infuenzae*	*Candida parapsilosis*
Staphylococcus lugdunensis	*Neisseria meningitidis*	*Candida tropicalis*
Staphylococcus spp.[b]	*Escherichia coli*	*Cryptococcus spp.*
Streptococcus agalactiae	*Enterobacter cloacae* complex	
Streptococcus pneumoniae	*Klebsiella pneuomniae* group	
Streptococcus pyogenes	Klebsiella aerogenes	
Streptococcus spp.[c]	Klebsiella oxytoca	
	Proteus spp.	
	Salmonella spp.	
	Serratia marcescens	
	Order Enterobacterales[e]	

[a]Gram positive organisms not identified by BCID2 include, but are not limited to, *Micrococcus, Peptostreptococcus, Clostridia spp., Bacillus spp. Paenibacillus spp.*
[b]*Staphylococcus spp.* includes, but is not limited to, *S. capitis, S. hominis, S. saprophyticus*
[c]*Streptococcus spp.* includes, but is not limited to, *S. mitis/oralis, S. gordonii, S. bovis, S. salivarius,* non-Group A/non-Group B β-hemolytic *Streptococcus*
[d][a]Gram negative organisms not identified by BCID2 include, but are not limited to, *Haemophilus*, *Moraxella*, *Achromobacter*, non-*baumanii Acinetobacter*, non-*aeruginosa Pseudomonas,* non-*fragilis Bacteroides* and other Gram-negative anaerobes. Piperacillin-tazobactam preferred as empiric treatment
[e]Order Enterobacterales includes, but is not limited to, *Citrobacter*¸ *Morganella, Providencia*, *Hafnia, Pantoea*

Last revised: 11/2025	**Antimicrobial Stewardship Subcommittee:** 11/2025	**P&T:** 01/2026

Revision History

12/2022	Revised gram-negative recommendations

The recommendations in this guide are meant to serve as treatment guidelines for use at Michigan Medicine facilities. If you are an individual experience a medical emergency, call 911 immediately. These guidelines should not replace a provider's profession medical advice based on clinical judgment, or be used in lieu of an Infectious Diseases consultation when necessary. As a result of ongoing research, practice guidelines may from time to time change. The authors of these guidelines have made all attempts to ensure the accuracy based on current information, however, due to ongoing research, users of these guidelines are strongly encouraged to confirm the information contained within them through and independent source.

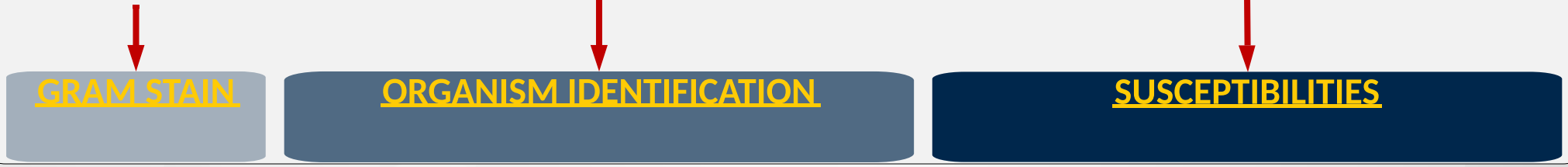
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How to use this guideline:

For patients with ONLY Gram stain results, refer to the left column (labeled GRAM STAIN) for treatment recommendations

For patients with organism identification results, refer to the middle column (labeled ORGANISM IDENTIFICATION) for treatment recommendations

For patients with antimicrobial susceptibility results, refer to the right column (labeled SUSCEPTIBILITIES) for treatment recommendations



GRAM STAIN

Yeast:
Micafungin

Consult ID

If suspicion for *Cryptococcus* or *Histoplasmosis* (fungemia in setting of pneumonia or meningitis in immunocompromised patient), call Infectious Diseases consult service for immediate antifungal recommendations

ORGANISM IDENTIFICATION

All Candida species (including C. auris):
Continue **Micafungin**

See Candidemia Guideline. Therapy should not be de-escalated until guideline criteria are met.

ID consult is strongly recommended.

If concern for urinary, ocular, endocarditis, or CNS infection, alternative therapy may be needed. Consult with ID

If Candida auris identified or prior colonization/infection with micafungin-resistant organism, immediately consult ID for treatment recommendations

Cryptococcus spp.:
Liposomal amphotericin B (Ambisome™)
+ **Flucytosine**

Consult ID

Histoplasma:
Liposomal amphotericin B (Ambisome™)

Consult ID

SUSCEPTIBILITIES

C. albicans, *C. parapsilosis*, *C. tropicalis*, *C. dublinensis*, and *C. lusitaniae*:
Consider de-escalation to **Fluconazole** for clinically stable patients with clearance of blood cultures and fluconazole susceptibility

Otherwise:
Micafungin

See Candidemia Guideline. Therapy should not be de-escalated until guideline criteria are met, in conjunction with ID consult recommendations

C. glabrata with fluconazole MIC ≤ 8 (SDD):
Consider de-escalation to **Fluconazole** for clinically stable patients with clearance of blood cultures

Otherwise:
Micafungin

Cryptococcus spp.:
Fluconazole may be appropriate for step down therapy when criteria is met in conjunction with ID consult recommendations

Histoplasma:
Step down therapy may be appropriate when clinically stable in conjunction with ID consult recommendations

GRAM STAIN

*Gram-positive cocci in clusters:
Vancomycin

ORGANISM IDENTIFICATION

S. aureus or *S. lugunensis* and *mecA/C + MREJ* negative (*S. aureus* only):
Endocarditis or CNS infection: **Oxacillin**
Other infections: **Cefazolin**

S. aureus or S. lugdunensis and *mecA/C + MREJ* positive or *mecA/C + MREJ* not performed:
Vancomycin
or
Daptomycin (if no pulmonary involvement)

Consult ID

Consider discontinuing adjunctive gram-negative therapy

Single positive culture for Coagulase-negative Staphylococcus or *S. epidermidis* in suspected infection of prosthetic material, neutropenia, or in hemodynamically unstable patients:

S. epidermidis and *mecA/C* negative:
Cefazolin

S. epidermidis and *mecA* positive or coagulase negative *Staphylococcus:*
Vancomycin

For patients who do not meet the above criteria, a single positive culture for coagulase-negative Staphylococcus or S. epidermidis may represent contamination, assess for possible source of infection and hold antibiotics if clinically stable

SUSCEPTIBILITIES

S. aureus or *S. lugdunensis* sensitive to methicillin:
Non-CNS/endocarditis: **Cefazolin** or **Oxacillin**
CNS infection or endocarditis: **Oxacillin**

S. aureus or *S. lugdunensis* intermediate or resistant to methicillin:
Vancomycin
or
Daptomycin (if no pulmonary involvement)

Antibiotic susceptibilities are only performed when coagulase-negative Staphylococcus or *S. epidermidis* grow from 2 or more bottles.

If growth from 1 blood culture bottle, assess for possible source of infection, repeat blood cultures, and hold antibiotics if clinically stable

Coagulase-negative *Staphylococcus* or *S. epidermidis* sensitive to methicillin:
Non-CNS/endocarditis: **Cefazolin**
CNS infection or endocarditis: **Oxacillin**

Coagulase-negative *Staphylococcus* or *S. epidermidis* intermediate or resistant to methicillin:
Vancomycin

GRAM STAIN

Gram-positive cocci in chains or pairs:
Vancomycin

Heme-onc, SICU, solid organ transplant:
Linezolid

Post-engraftment HCT recipients:
Linezolid

Pre-engraftment HCT recipients:
Daptomycin

ORGANISM IDENTIFICATION

E. faecalis and *vanA/vanB* Negative:
Ampicillin
(consider **piperacillin-tazobactam** as alternative for intra-abdominal infections)
Life-threatening PCN allergy: **Vancomycin**

E. faecalis and *vanA/vanB* positive:
Ampicillin
(consider **piperacillin-tazobactam** as alternative for intra-abdominal infections)
Life-threatening PCN allergy: **Linezolid** or **Daptomycin** for pre-engraftment HCT patients

E. faecium and *vanA/vanB* negative:
Vancomycin

E. faecium and *vanA/vanB* positive:
Linezolid or
Daptomycin for BMT pre-engraftment HCT patients

E. casseliflavus, E. gallinarium:
Linezolid or
Daptomycin for BMT pre-engraftment HCT patients

Other *Enterococcus* species:
Vancomycin

S. pneumoniae, S. anginosus or *Streptococcus* species:
Non-CNS/endocarditis: **Ceftriaxone**
CNS infection: **Ceftriaxone + Vancomycin**
Endocarditis or Febrile neutropenia: **Vancomycin**

S. agalactiae or *S. pyogenes*:
Penicillin or **Ampicillin**

SUSCEPTIBILITIES

Penicillin-based antibiotics should be first line therapy for all *Enterococcus* species if sensitive:
Ampicillin
(consider **ampicillin-sulbactam** or **piperacillin-tazobactam** for intra-abdominal infections)

Life-threatening PCN allergy or ampicillin-resistant *Enterococcus*:
Vancomycin

Patients with vancomycin allergy or ampicillin and vancomycin-resistant *Enterococcus*:
Linezolid or
Daptomycin for BMT pre-engraftment HCT patients

Patients with suspected endocarditis will likely require combination therapy and ID consult is strongly recommended

Penicillin-based antibiotics should be first line therapy for all *Streptococcus* species infections, if sensitive:
Penicillin or **Ampicillin**

GRAM STAIN

Gram-negative bacilli::
Piperacillin-tazobactam
or
Cefepime
(add **metronidazole** for intra-abdominal infections)

*Evaluate if patient has history of resistance to piperacillin-tazobactam or cefepime with prior year and modify therapy accordingly

ORGANISM IDENTIFICATION

E. coli, K. pneumoniae, K. oxytoca, Proteus, Serratia, Morganella, Salmonella, Enterobacterales:
No *CTX-M*, *KPC*, *IMP*, *VIM*, *NDM*, *OXA* detected:
Cefepime or **Piperacillin-tazobactam**

CTX-M positive:
Meropenem

KPC positive:
Meropenem-vaborbactam

IMP, *VIM*, or *NDM* (non-*E. coli)* positive:
Aztreonam-avibactam

NDM positive *E. coli* only:
Aztreonam-avibactam + tigecycline

OXA positive:
Ceftazidime-avibactam

Enterobacter cloacae complex, Citrobacter freundii, or Klebsiella aerogenes:
No *CTX-M*, *KPC*, *IMP*, *VIM*, *NDM*, *OXA* detected:
Cefepime

CTX-M positive:
Meropenem

KPC positive:
Meropenem-vaborbactam

IMP, *VIM*, or *NDM* positive:
Aztreonam-avibactam or **Cefiderocol**

OXA positive:
Ceftazidime-avibactam

SUSCEPTIBILITIES

Narrow antibiotic selection based on susceptibility results, clinical status, concomitant infections:

- Narrow-spectrum antibiotics are preferred if no resistance or allergies. These include **ampicillin**, **penicillin**, **ampicillin-sulbactam**, **cefazolin**, and **cefuroxime**.
- ID consult is strongly encouraged for patients with infections from organisms with *KPC*, *IMP*, *VIM*, *NDM*, or *OXA* resistance genes
- *Serratia* is at low risk for producing an inducible AmpC ß-lactamase but caution should still be used when using ceftriaxone or piperacillin-tazobactam as definitive therapy in infections with a high organism burden or uncontrolled source (e.g., pneumonia, meningitis). Consider cefepime, a carbapenem, or non-ß-lactam as approrpriate.

Narrow antibiotic selection based on susceptibility results, clinical status, concomitant infections:

- ID consult is strongly encouraged for patients with infections from organisms with *KPC*, *IMP*, *VIM*, *NDM*, or *OXA* resistance genes

- *Enterobacter cloacae*, *Citrobacter freundii*, *and Klebsiella aerogenes* frequently have an inducible ß-lactamase resistance gene (*AmpC*), which can confer resistance to penicillin, ampicillin, ampicillin/sulbactam, and 1st-3rd generation cephalosporins. **Cefepime** should be first-line therapy if susceptible.

- *Citrobacter koseri* is not associated with having *AmpC* gene, and narrow spectrum antibiotics should be prescribed if susceptible.

GRAM STAIN

Gram-negative bacilli::
Piperacillin-tazobactam
or
Cefepime
(add **metronidazole** for intra-abdominal infections)

*Evaluate if patient has history of resistance to cefepime with prior year and modify therapy accordingly

ORGANISM IDENTIFICATION

Pseudomonas aeruginosa:
No *IMP, VIM, NDM* detected:
Cefepime or **Piperacillin-tazobactam.** Consider empiric double coverage with tobramycin

IMP, VIM, or *NDM* positive:
Cefiderocol + tobramycin until susceptibilities result

CTX-M, KPC, OXA positive:
Contact ID – unusual genotype

Acinetobacter baumanii:
No *IMP, VIM, NDM, OXA* detected:
Sulbactam-durlobactam

IMP, VIM, or *NDM* positive:
Cefiderocol + **minocycline** until susceptibilities result

CTX-M or KPC positive:
Sulbactam-durlobactam
Contact ID – unusual genotype

SUSCEPTIBILITIES

Narrow antibiotic selection based on susceptibility results, clinical status, concomitant infections.

- If Pseudomonas isolate is resistant to **ceftolozane-tazobactam**, **ceftazidime-avibactam**, and **meropenem-vaborbactam,** request **cefiderocol** susceptibilities from microbiology lab (phone number 6-6831)
- Double coverage of *Pseudomonas* is not generally indicated after susceptibilities are available, unless isolate is resistant to all ß-lactam antibiotics, cystic fibrosis patient, or decompensating on susceptible antibiotics. Please call ID if this is the case.
- non-*baumannii Acinetobacter spp.* (e.g., *iwofii*, *radioresistans*, etc.) will initially appear as Gram negative rod with a negative Biofire panel

Narrow antibiotic selection based on susceptibility results, clinical status, concomitant infections.

- There is no evidence double coverage of Acinetobacter improves outcomes. The decision to double cover should be made based on source of bacteremia, severity of infection, and patient's medical history.
- Consider definitive combination therapy for carbapenem-resistant *A. baumannii* based on susceptibility information and infectious diseases consultation
- non-*baumannii Acinetobacter spp.* (e.g., *iwofii*, *radioresistans*, etc.) will initially appear as Gram negative rod with a negative BIofire panel and should be treated empirically with cefepime or piperacillin-tazobactam

GRAM STAIN

*Gram-negative bacilli:: **Piperacillin-tazobactam** or **Cefepime**

*Evaluate if patient has history of resistance to cefepime with prior year and modify therapy accordingly

Gram-positive rod: Most likely the result of skin flora contamination of blood culture

Consider treatment in HD unstable, prosthetic material with suspected infection, BMT, Neutropenia: **Vancomycin**

If concern for *Listeria*: **Ampicillin**

ORGANISM IDENTIFICATION

Achromobacter: **Piperacillin-tazobactam**
Life-threatening PCN allergy: **Meropenem**

(Avoid cefepime unless susceptibility is verified)

Stenotrophomonas maltophilia:
non-neutropenic or critically ill due to *S. maltophilia*: **Trimethoprim-sulfamethoxazole**
Sulfa-allergy: **Levofloxacin** + **minocycline**

Neutropenic or critically ill due to *S. maltophilia*: **Cefiderocol ± minocycline**

Bacillus, *Lactobacillus*, and *Corynebacterium* spp. are possible contaminants, consider treatment in HD unstable, prosthetic material with suspected infection, BMT, solid organ transplant, neutropenia

Bacillus or *Corynebacterium* spp.: **Vancomycin**

Lactobacillus: **Piperacillin-tazobactam**

Listeria: **Ampicillin**

SUSCEPTIBILITIES

Narrow antibiotic selection based on susceptibility results, clinical status, concomitant infections.

- *Achromobacter* is frequently multi-drug resistant, and ID consult is encouraged to guide appropriate management of these infections
- Piperacillin-tazobactam and cefepime do not have activity against Stenotrophomonas
- Request **cefiderocol** susceptibility testing from the clinical microbiology laboratory

Narrow antibiotic selection based on susceptibility results, clinical status, concomitant infections.

- Susceptibilities will not be routinely performed by the microbiology lab. Please call to request susceptibilities if strong suspicion for infection
- Daptomycin should not be used as definitive therapy for *C. striatum* due to rapid development of resistance. It should be used with caution in all other Gram positive rods